Path Codes on Shared Knowledge Graphs

Part 5: Progressive Transmission over Noisy Channels

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Abstract

On a hierarchy, the first decisions of a path code name a coarse class of the meaning. If each level is protected by its own channel code and the receiver stops at the first block that fails, a channel failure costs resolution instead of producing a wrong meaning. We give the expected semantic distortion of this scheme exactly (Theorem 1), show that the best split of a fixed channel budget among the levels is found exactly by dynamic programming (Proposition 2), and identify what progressivity costs in rate: identifying classes exactly costs $H(C_j) - R(\bar{d}_j)$ bits at level j, which is zero at the first and last levels and strictly positive at every non-trivial intermediate level, because no deterministic test channel is optimal (Lemma 3, Proposition 4). Against separate source and channel coding, layering gains nothing when the channel is known and the blocks are long. It gains when the transmitter does not know the channel: coarse levels survive below the design signal-to-noise ratio, at a price in channel uses that we compute (Proposition 5). Numerically, giving more protection to coarser levels is optimal in only some cases, so it is a heuristic and not a law.

1 Setting

Meanings $M \in \mathcal{M}$ have law P_M and distortion d on a concept graph (Part 3). A *hierarchy of height* h is a chain of partitions $\mathcal{Q}_0 \sqsupseteq \mathcal{Q}_1 \sqsupseteq \cdots \sqsupseteq \mathcal{Q}_h$ of $\mathcal{M}$, with $\mathcal{Q}_0 = \{\mathcal{M}\}$ and $\mathcal{Q}_h$ the partition into singletons; $C_j = c_j(M)$ is the class of M in $\mathcal{Q}_j$. The best estimate of M from C_j has expected distortion

$$\bar{d}_j = \mathbb{E}\Big[\min_{\hat{m}\in\mathcal{M}} \mathbb{E}\big[d(M,\hat{m}) \mid C_j\big]\Big] = \sum_{B\in\mathcal{Q}_j} \min_{b} \sum_{m\in B} P_M(m)\, d(m,b),$$

the distortion $D_{\mathcal{Q}_j}$ of the quotient code of $\mathcal{Q}_j$ when representatives may lie anywhere (Part 3 restricts $b \in B$; the two coincide for the ultrametric of a tree, where the best representative of a class lies in it). Since $\mathcal{Q}_j$ refines $\mathcal{Q}_{j-1}$, $\bar{d}_0 \geq \bar{d}_1 \geq \cdots \geq \bar{d}_h = 0$.

By Corollary 3 of Part 3, an adapted path code that selects root-to-leaf branches of the hierarchy identifies C_j with its first j decisions. We call the description of C_j given C_{j-1} the *level-j block*.

Definition 1 (Layered transmission)**.** Level j is described with k_j source bits and protected by its own channel code of n_j uses of a memoryless channel W, with $\sum_j n_j = n$. The receiver decodes the blocks in order and stops at the first block it declares failed. Let ε_j be the probability that block j fails and J the index of the first failed block ($J = h+1$ if none fails). The receiver outputs the best estimate of M given C_{J-1}.

The assumptions are: (L1) the channel noise is independent of the source and of the other blocks, which holds when the blocks use disjoint channel uses of a memoryless channel; (L2) failures are detected, as on an erasure channel or with an error-detecting code (Remark 2 removes this assumption at a stated cost).

2 Expected semantic distortion

Theorem 1 (Expected distortion of layered transmission)**.** *Under (L1)–(L2),*

$$\mathbb{E}[d(M,\hat{M})] = \sum_{j=0}^{h} \Pr[J = j+1]\, \bar{d}_j, \qquad \Pr[J = j+1] = \varepsilon_{j+1} \prod_{i=1}^{j} (1-\varepsilon_i), \tag{1}$$

with $\varepsilon_{h+1} := 1$.

Proof. By (L2) every block accepted before J is correct, so on $\{J = j+1\}$ the receiver knows C_j. By (L1) the event $\{J = j+1\}$ is independent of M, so the conditional law of M given C_j and $\{J = j+1\}$ is the law given C_j, and the best estimate has expected distortion $\bar{d}_j$. The probability of $\{J = j+1\}$ is the product of the success probabilities of blocks $1, \ldots, j$ and the failure probability of block $j+1$, again by (L1). Summing over j gives (1). □

Remark 1 (What the theorem compares). (a) *Index code with uniform protection.* All $k = \sum_j k_j$ bits in one block of n uses fails with probability ε_U; with detection, the receiver falls back to the global estimate, so $\mathbb{E}[d] = \varepsilon_U\, \bar{d}_0$. Without detection a failure yields an arbitrary meaning. (b) *Embedded index code.* A code designed level by level on a chain of partitions obeys (1) with that chain's $\bar{d}_j$. On a tree the path code *is* such a code, with the chain of ancestors; what the graph supplies is the chain itself, without design, and, on graphs with synonymy, a chain whenever the Viterbi selection forms a tree (Part 1). Any advantage of the path code must therefore come from the chain, never from the layering: this is the comparison that Section 4 quantifies. (c) The sequential decoder ignores blocks after the first failure. A decoder that uses them can only do better, so (1) is achievable and is an upper bound on the optimal decoder for the same codes.

Remark 2 (Undetected errors). If block j is accepted in error with probability η_j, the receiver may output a wrong class. Each such event costs at most $d_{\max} = \max_{m,\hat{m}} d(m,\hat{m})$, so the expected distortion is at most (1) plus $d_{\max} \sum_j \eta_j$. A cyclic redundancy check of c bits on a block makes η_j of order 2^{-c} for long blocks; Section VIII of the manuscript reports the value measured with the practical code.

3 Allocation of a channel budget

Proposition 2 (Exact allocation)**.** *Let* $\varepsilon(n_j, k_j)$ *be any function, with* $\varepsilon(0,k) = 1$ *for* $k > 0$. *Define* $V_{h+1}(r) = \bar{d}_h$ *and, for* $j = h, \ldots, 1$,

$$V_j(r) = \min_{0 \le n_j \le r} \Big[\varepsilon(n_j,k_j)\, \bar{d}_{j-1} + \big(1 - \varepsilon(n_j,k_j)\big)\, V_{j+1}(r-n_j) \Big].$$

Then $V_1(n)$ *is the minimum of* (1) *over integer allocations with* $\sum_j n_j = n$, *the minimizers are recovered by backtracking, and the computation takes* $O(hn^2)$ *evaluations of* ε.

Proof. Write (1) as $\varepsilon_1 \bar{d}_0 + (1-\varepsilon_1)\big[\varepsilon_2 \bar{d}_1 + (1-\varepsilon_2)[\cdots]\big]$. The bracket after level j depends only on $(n_{j+1}, \ldots, n_h)$, which must sum to $r - n_j$ when r uses remain before level j. Since $1 - \varepsilon_j \ge 0$, minimizing the outer expression requires minimizing the bracket for each n_j; this is $V_{j+1}(r-n_j)$ by induction on j from h down to 1. There are h levels, $n+1$ values of r and at most $n+1$ choices of n_j. □

Observation 1 (Coarser levels do not always deserve more protection). With the normal approximation of the block error probability on a binary symmetric channel, $\varepsilon(n,k) \approx Q\big((nC-k)/\sqrt{nV}\big)$ [1], the optimal code rates k_j/n_j were non-decreasing in j (more protection for coarser levels) in only 26 of 40 random hierarchies (`check_t34.py`, part 2). When a coarse level carries few bits and a finer level reduces distortion a lot, the program prefers to protect the finer level. The usual rule of unequal error protection is therefore a heuristic here; Proposition 2 replaces it.

4 What progressivity costs in rate

Consider now long blocks of independent meanings, so that rates are per meaning and source coding is ideal. A progressive code (Definition 8 of the manuscript) must let a prefix determine C_j; its prefix rate is therefore at least $H(C_j)$. An unconstrained code that only needs distortion $\bar{d}_j$ at that point needs $R(\bar{d}_j)$.

Lemma 3 (Deterministic test channels are not optimal). *Let $d(m,\hat{m}) < \infty$ for all $m, \hat{m}$, and let $0 < D < D_{\max}$. No test channel that is a deterministic map $m \mapsto b(m)$ taking at least two values attains $R(D)$ at distortion D.*

Proof. Suppose such a map b is optimal at D. Because $D < D_{\max}$, $R(D) > 0$ and the distortion constraint is active; the problem is convex and $D > 0$ makes it strictly feasible, so by Lagrangian duality there is $\beta > 0$ such that the channel of b minimizes $I(M;\hat{M}) + \beta\,\mathbb{E}\,d(M,\hat{M})$ over all channels. Every minimizer of this Lagrangian has the form $W(\hat{m} \mid m) = q(\hat{m})\,2^{-\beta d(m,\hat{m})} / \sum_{\hat{m}'} q(\hat{m}')\,2^{-\beta d(m,\hat{m}')}$, where q is its output law [3, Ch. 10]. The output law of b puts positive mass on at least two values $b_1 \neq b_2$. Take m with $b(m) = b_1$: the deterministic channel gives $W(b_2 \mid m) = 0$, whereas the form above gives $W(b_2 \mid m) > 0$ because $q(b_2) > 0$ and $d(m,b_2) < \infty$. This is a contradiction. □

Proposition 4 (Cost of identifying classes). *Let $\Delta_j = H(C_j) - R(\bar{d}_j)$, and let $d(m,\hat{m}) > 0$ for $m \neq \hat{m}$. (a) $\Delta_j \geq 0$ for every j. (b) $\Delta_0 = \Delta_h = 0$. (c) The total rate of the layered description is $\sum_j H(C_j \mid C_{j-1}) = H(M)$: progressivity costs nothing at the last level. (d) $\Delta_j > 0$ for every level $\mathcal{Q}_j$ that is neither the trivial partition nor the partition into singletons.*

Proof. Let $\hat{M} = b(C_j)$ be the best representative of the class, which may be shared by several classes. (a) This deterministic test channel has distortion $\bar{d}_j$ and rate $H(\hat{M}) \leq H(C_j)$, so $R(\bar{d}_j) \leq H(C_j)$. (b) $H(C_0) = 0 = R(\bar{d}_0)$ because $\bar{d}_0 = D_{\max}$; at $j = h$, $\bar{d}_h = 0$ and $R(0) = H(M) = H(C_h)$ because d is positive off the diagonal. (c) Chain rule, since C_{j-1} is a function of C_j. (d) Since $\mathcal{Q}_j$ is not the partition into singletons, some class has two meanings, one of which is not its own representative, so $\bar{d}_j > 0$. If $\bar{d}_j = D_{\max}$, then $R(\bar{d}_j) = 0 < H(C_j)$ because $\mathcal{Q}_j$ is not trivial. Otherwise $0 < \bar{d}_j < D_{\max}$; then $\hat{M}$ takes at least two values (a single value b would give distortion $\mathbb{E}\,d(M,b) \geq D_{\max}$), and Lemma 3 gives $R(\bar{d}_j) < H(\hat{M}) \leq H(C_j)$. □

Proposition 4(d) strengthens Theorem 1(c) of Part 3 (and Theorem 4 of the manuscript), which exhibited one example of a strictly suboptimal quotient: *every* non-trivial quotient code is strictly suboptimal. Numerically, `check_t34.py` (part 3) finds $\Delta_j > 0$ in all 18 non-trivial levels of random height-two trees, between 0.0025 and 0.409 bits.

Remark 3 (The cost is in identifying classes, not in being progressive). If the Shannon-optimal test channels at the successive distortions form a Markov chain, the source is successively refinable [4] and a progressive *lossy* code attains $R(\bar{d}_j)$ at every level. The loss Δ_j comes from requiring that a prefix identify a class exactly, which is what a path code delivers and what a receiver that must act on "floor two" needs. By Proposition 4(d) this loss is strictly positive at every non-trivial level,

even when the hierarchy is ultrametric and the source is successively refinable: it is the rate price of a semantic guarantee, not of progressivity.

5 Layering versus separation

Consider B independent meanings, coded jointly at each level: level j carries about $B\,H(C_j \mid C_{j-1})$ bits, losslessly given the classes already decoded, over $B\rho_j$ uses of the channel, so the total budget is $\rho = \sum_j \rho_j$ uses per meaning. A uniform scheme sends all $B\,H(M)$ bits in one block of $B\rho$ uses. Let $C(\gamma)$ be the capacity at signal-to-noise ratio γ, increasing in γ.

Proposition 5 (When layering helps)**.** *Let $B \to \infty$ with ρ_j fixed, and use channel codes that achieve capacity. (a)* Known channel. *At a known γ, both schemes deliver every meaning exactly if and only if their channel budget is at least $H(M)/C(\gamma)$ per meaning; layering neither gains nor loses. (b)* Unknown channel. *Choose target ratios $\gamma_{(1)} \le \cdots \le \gamma_{(h)}$ and $\rho_j = H(C_j \mid C_{j-1})/C(\gamma_{(j)})$. At operating ratio γ the layered scheme attains $\mathbb{E}[d] \to \bar{d}_{J(\gamma)}$ with $J(\gamma) = \max\{j : \gamma_{(i)} < \gamma \ \forall i \le j\}$, while the uniform scheme designed for $\gamma_{(h)}$ attains $\bar{d}_0$ for every $\gamma < \gamma_{(h)}$. The layered scheme uses $\sum_j H(C_j \mid C_{j-1})/C(\gamma_{(j)}) \ge H(M)/C(\gamma_{(h)})$ channel uses per meaning, with equality if and only if all targets coincide.*

Proof. By the channel coding theorem, a block of rate below $C(\gamma)$ fails with probability tending to zero; by the strong converse, made quantitative by the normal approximation of [1], a block of rate above $C(\gamma)$ fails with probability tending to one. (a) Layered: every level is delivered if and only if $H(C_j \mid C_{j-1}) < \rho_j C(\gamma)$ for all j, which needs $\rho > H(M)/C(\gamma)$ by summing; uniform: $H(M) < \rho C(\gamma)$; both thresholds coincide. (b) Level j succeeds if and only if $\gamma > \gamma_{(j)}$; the first failing level is $J(\gamma) + 1$, and Theorem 1 with $\varepsilon_j \in \{0, 1\}$ gives $\bar{d}_{J(\gamma)}$. The uniform block has rate $C(\gamma_{(h)})$ and fails for every $\gamma < \gamma_{(h)}$. The budget inequality follows from $C(\gamma_{(j)}) \le C(\gamma_{(h)})$. □

Remark 4 (What this does not claim). Layering is time sharing among levels. On a degraded broadcast channel, superposition coding achieves more [5], and joint source–channel schemes degrade gracefully as well [6, 7]; the layered path code is a simple achievable scheme, not an optimal one. In the finite-blocklength regime and at a known channel, splitting a budget into shorter blocks raises each block's error probability [1, 2], so layering can lose; the manuscript measures this numerically (E5) instead of claiming a sign.

Example (`check_t34.py`, part 4). A random tree of height three with Dirichlet source has $H(C_j \mid C_{j-1}) = (0.353, 0.053, 0.828)$ and $H(M) = 1.234$ bits. With BPSK and hard decisions, targets of 0, 3 and 6 dB need 1.497 channel uses per meaning, 18.4% more than the 1.265 of a uniform code designed for 6 dB. Below 6 dB the uniform code delivers only the global estimate ($\bar{d}_0 = 0.391$), while the layered one keeps the first level above 0 dB ($\bar{d}_1 = 0.223$) and the first two above 3 dB ($\bar{d}_2 = 0.191$).

Design implication (RTLS). Tags far from an anchor see a lower signal-to-noise ratio. Layering lets every tag deliver the floor, the tags with a fair link also the wing, and only those with a good link the room, at a price in airtime that Proposition 5 gives in closed form.

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